ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a single kind of particle with fixed composition	Pure substances have fixed composition.
2	**(b)** a heterogeneous mixture whose particles settle on standing	Large particles.
3	**(b)** pencil does not dissolve in the solvent	Ink would move.
4	**(c)** distilled water	Single compound.
5	**(c)** fog	Water droplets in air.
6	**(d)** moves farther up the paper	Greater solubility.
7	**(c)** brass	Alloy has uniform composition.
8	**(b)** smaller than the pores of the paper	Colloidal particles are small.
9	**(d)** homogeneous mixtures of metals	Metal solutions.
10	**(c)** heterogeneous	Components are visible.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Homogeneous: uniform, e.g. salt solution (1); Heterogeneous: non-uniform, e.g. sand and water (1).	1+1
12	Separation by spinning at a high speed (1); Cream from milk, blood cells from plasma (1).	1+1
13	Mass of solution = 100 g (1); 25 per cent (1).	1+1
14	Maximum mass of solute that dissolves in 100 g of solvent at a given temperature (1); Generally increases with temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heat the mixture: ammonium chloride sublimes (1); Add water to the residue, filter sand (1); Evaporate the filtrate to get salt (1).	1+1+1
16	12/100 x 250 (1); = 30 g (1); Water = 220 g (1).	1+1+1
17	Saturated: no more solute dissolves (1); Unsaturated: can dissolve more (1); Heat or add solvent to make unsaturated (1).	1+1+1
	OR: Dissolve in hot water to a saturated solution and filter (1); Cool slowly (1); Pure crystals form and are dried (1).	1+1+1
18	Alcohol = 15/100 x 200 = 30 mL (1); Add to some water in a measuring cylinder (1); Make up to 200 mL (1).	1+1+1
	OR: Impurities would stay with crystals (1); Crystals should form on cooling (1); Water of crystallisation could be lost (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pure: fixed composition (1); Mixture: variable (1); Gold: pure; air, sugar solution, brass: homogeneous (1); Milk: colloid, treated as heterogeneous (1); Properties differ (1).	1+1+1+1+1
	OR: Spot above the baseline (1); Solvent rises by capillary action (1); Components move at different rates (1); Compare heights (1); Used for dyes and pigments (1).	1+1+1+1+1
20	Dissolve salt, mix chalk and milk in water (1); Observe clarity (1); Filter each (1); Shine light (1); Observe settling (1).	1+1+1+1+1
	OR: Saturated solution is cooled slowly (1); Crystals form (1); Impurities remain in the liquid (1); Sugar and alum (1); Purified solids obtained (1).	1+1+1+1+1

Mistakes to avoid:

- Calculating mass per cent using the mass of the *solvent* instead of the mass of the *solution* (solute + solvent).
- Saying that a clear liquid must be a pure substance; a clear solution such as salt water is a mixture.
- Using filtration to separate a dissolved solid; filtration works only for insoluble particles, not for a solute.

- Confusing evaporation (solvent lost, solid left) with distillation (solvent collected by condensing); and using fractional distillation when the boiling points differ by more than 25 K.
- Thinking that the solubility of a gas also increases when the temperature rises; the solubility of gases in water decreases.
- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** centrifugation	Spinning.
2	**(b)** filtration	Insoluble solid.
3	**(d)** solvent	Dissolving medium.
4	**(c)** sublimation	Ammonium chloride sublimes.
5	**(a)** evaporation (and condensation)	Dissolved solute.
6	**(c)** saturated	At its solubility limit.
7	**(b)** distillation	Boiling points differ.
8	**(b)** husk from grain using wind	Density difference.
9	**(c)** 10 per cent	10/100 x 100.
10	**(c)** crude oil into petrol, kerosene, etc.	Close boiling points.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Boiling a liquid and condensing its vapour (1); Water from salt solution or miscible liquids (1).	1+1
12	Milk: liquid in liquid (1); Fog: liquid in gas (1).	1+1
13	Scattering of light by colloidal particles (1); Beam of sunlight through a dusty room or milk (1).	1+1
14	Pencil does not dissolve in the solvent (1); Ink would spread and interfere (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Air is cooled and compressed to a liquid (1); Liquid is warmed slowly (1); Nitrogen (lowest b.p.), argon, oxygen come out in this order (1).	1+1+1
16	Spot of ink near the bottom of a strip (1); Dip the end in the solvent (1); Dyes rise to different heights and separate (1).	1+1+1
17	Higher temperature gives faster evaporation (1); Wind removes vapour (1); More surface area exposed (1).	1+1+1
	OR: Pour the mixture in the funnel (1); Allow layers to settle (1); Open the tap to run out the denser layer (1).	1+1+1
18	(a) Particle size difference (1); (b) Boiling point difference (1); (c) Only one component sublimes (1).	1+1+1
	OR: Stronger and resistant to corrosion (1); Brass: fittings (1); Steel: construction; bronze: statues (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Flask, condenser, receiver (1); Simple: salt from water (1); Fractional: column for close boiling points (1); Crude oil refining (1); Air separation (1).	1+1+1+1+1
	OR: Air is compressed and cooled (1); Liquid air warmed slowly (1); Nitrogen about -196 degrees Celsius (1); Argon about -186 degrees Celsius (1); Oxygen about -183 degrees Celsius (1).	1+1+1+1+1
20	(a) Filtration (1); (b) Centrifugation (1); (c) Magnet (1); (d) Evaporation (1); (e) Sublimation (1).	1+1+1+1+1
	OR: Sea salt from sea water (1); Petrol from crude oil (1); Cream from milk (1); Purified water (1); Metals from ores (1).	1+1+1+1+1

Mistakes to avoid:

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- Forgetting that a colloid does not settle and cannot be filtered, but still shows the Tyndall effect.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Exploring Mixtures and Their Separation** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** iron filings from sulphur	Magnetic material.
2	**(b)** 25 per cent	20/80 x 100.
3	**(d)** gives pure crystals and does not evaporate to dryness	Slow cooling gives crystals.
4	**(c)** increases with temperature	Warm water dissolves more.
5	**(a)** smaller than 1 nm and do not scatter light	True solution.
6	**(a)** a separating funnel	Layers separate.
7	**(b)** sedimentation, decantation and filtration	Water treatment.
8	**(b)** milk	Colloid scatters light.
9	**(b)** colour and writing	chroma and graphein.
10	**(c)** fractional distillation of liquid air	Different boiling points.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Direct change of a solid into vapour (1); Camphor, ammonium chloride, naphthalene (1).	1+1
12	Solution is clear and stable; suspension is turbid and settles (1); Solution passes through filter paper; suspension is filtered out (1).	1+1
13	Homogeneous mixture of metals or of a metal and a non-metal (1); Brass: copper and zinc (1).	1+1
14	To obtain pure substances (1); e.g. salt from sea water, petrol from crude oil (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Solution: <1 nm, stable, no effect (1); Colloid: 1-1000 nm, stable, shows effect (1); Suspension: >1000 nm, unstable, scatters but settles (1).	1+1+1
16	Particles are between 1 and 1000 nm (1); Show the Tyndall effect (1); Do not settle or filter out (1).	1+1+1
17	Oil molecules are unlike water molecules (1); Alcohol molecules interact with water (1); Like dissolves like (1).	1+1+1
	OR: Mass of solution = 136 g (1); 36/136 x 100 (1); = 26.5 per cent (1).	1+1+1
18	Sedimentation (1); Filtration through sand beds (1); Chlorination to kill germs (1).	1+1+1
	OR: Pass a beam of light through the mixture in a dark room (1); Colloid scatters the beam making it visible (1); True solution shows no path (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	m/m = mass of solute/mass of solution x 100 (1); v/v = volume of solute/volume of solution x 100 (1); m/v = mass of solute/volume of solution x 100 (1); Example 10 g in 90 g gives 10 per cent (1); Example 5 mL in 100 mL gives 5 per cent (1).	1+1+1+1+1
	OR: Boil starch with water and cool (1); Looks translucent (1); Passes through filter paper (1); Shows Tyndall effect (1); Does not settle (1).	1+1+1+1+1
20	Solids: generally increases with temperature (1); KNO3 rises sharply (1); NaCl rises slowly (1); Gases decrease with temperature (1); Like dissolves like (1).	1+1+1+1+1
	OR: Suspension by filtration or settling (1); Colloid by centrifugation (1); Coagulation by alum (1); Dialysis uses semi-permeable membrane (1); Examples (1).	1+1+1+1+1

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